

Unidades

$$\rho = \frac{\text{kg}}{\text{m}^3} \quad \mu = \frac{\text{Pa} \cdot \text{s}}{\frac{\text{N}}{\text{m}^2}} \quad \text{s}$$

$$P = \frac{\text{Pa} \cdot \text{N}}{\frac{\text{m}^2}} \quad \text{T (Tensión Superficial)} = \frac{\text{N}}{\text{m}}$$

$$\bar{E} \text{ (Tensión de Corte)} = \frac{\text{N}}{\text{m}^2}$$

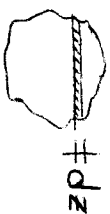
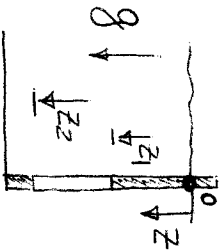
Formulas

$$\bar{E} = \mu \frac{dV}{dy}$$

$$P = \rho g h + P_0$$

Compuertas

Planas



$$F = \int_{z_1}^{z_2} \rho g z da$$

$$M^0 = \int_{z_1}^{z_2} \rho g z^2 da$$

$$\Rightarrow Z_C = \frac{M^0}{F}$$

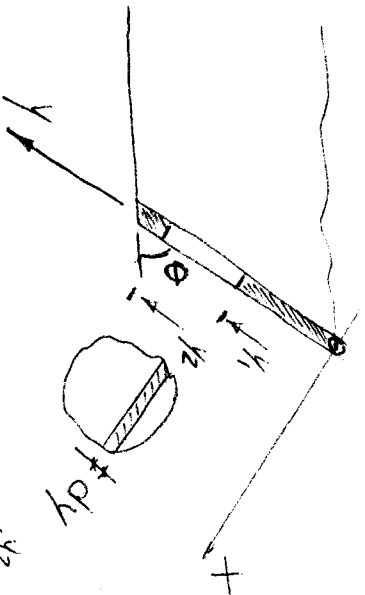
$$F = \rho g S_0$$

$$M^0 = \rho g I_{\infty}$$

$$S_0 = A \cdot Z_G$$

$$I_{\infty} = I_G + Z_G^2 \cdot A$$

Inclinadas



$$F = \int_{y_1}^{y_2} \rho g \cdot y da \cdot \sin \theta$$

$$M^0 = \int_{y_1}^{y_2} \rho g y^2 da \sin \theta \Rightarrow Y_C = \frac{M^0}{F}$$

$$F = \rho g \sin \theta \cdot S_0$$

$$M^0 = \rho g \sin \theta \cdot I_{\infty}$$

Bernoulli

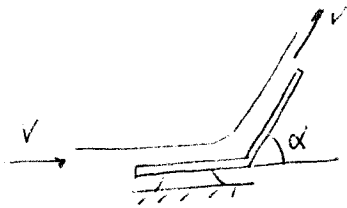
$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$Q_1 = Q_2 \quad (\text{continuidad})$$

Empuje

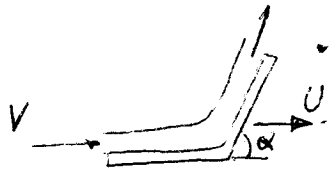
$$\bar{E} = -\frac{\partial}{\partial t} \int_{V_c} \rho \bar{V} dV - \int_{A_c} \rho \cdot \bar{V}_r \cdot (\bar{V}_r \cdot d\bar{A}) - \int_{A_c} P_r d\bar{A} - \bar{F}_{fu} - \int_{V_c} \rho \cdot \bar{\omega} \cdot dV$$

Álabes Fijos



$$\bar{E} = \bar{V}^2 \cdot f \cdot A (1 - \cos \alpha)$$

Álabes Móviles



$$\bar{E} = (V - U)^2 \cdot f \cdot A (1 - \cos \alpha)$$

Rueda Pelton

$$\bar{E} = \bar{V} \cdot f \cdot A \cdot (\bar{V} - \bar{U}) (1 - \cos \alpha)$$

$$U = W \cdot r_{medio}$$

$$P_{\text{útil}} = \bar{E} \cdot \bar{U}$$